Chapter 3

The fuel, air and flue-gas circuits

Having looked at the steam and water circuits of boilers and HRSGs, we now move on to examine the plant which is involved in the combustion of fuel in boilers.

The heat used for generating the steam is obtained by burning fuel in a furnace, or combustion chamber, but to do this requires the provision of air which is provided by a forced-draught (FD) fan (in larger boilers, two such fans are provided). After the fuel has been burned, the hot products of combustion are extracted from the furnace by another fan, the induced-draught (ID) fan, and fed to the chimney. Again, two ID fans are provided on larger boilers.

In this chapter we shall examine not only the burners or other equipment used to burn the fuel but also the fans and air heaters. Finally we shall briefly examine how gas turbines are used in combined-cycle plant.

3.1 The furnace

In boiler plant the heat used for boiling the water is obtained by burning a fossil fuel (unlike the HRSG, where the heat is delivered by the exhaust of a gas turbine). This process of combustion is carried out in the furnace, and comprises a chemical reaction between the combustible material and oxygen. If insufficient oxygen is available some of the combustibles will not burn, which is clearly inefficient and polluting. On the other hand, the provision of too much oxygen leads to inefficient operation and to corrosion and undesirable emissions from the stack due to the combination of the surplus oxygen with other components of the flue gases.

The oxygen for combustion is provided in air, which contains around 21% of the gas. However, air also contains around 77% nitrogen, and the combustion process results in the production of nitrogen dioxide (NO_2) and nitric oxide (NO). These gases (plus nitrous oxide, N_2O) are collectively called nitrogen oxides, or NO_x for short, and because they are often blamed for various detrimental effects on the environment a high level of attention must be given to minimising their production.

Unfortunately, high combustion efficiencies invariably correspond with the production of high levels of nitrogen oxides, and therefore NO_x reduction involves careful design of the burners so as to yield adequate combustion efficiency with minimal smoke and carbon monoxide generation.

3.1.1 Firing arrangements

The combustion of oil, gas or pulverised coal is performed in burners. These may be arranged on one wall of the combustion chamber (which is therefore called 'front-fired'), or on facing walls ('opposed fired') or at the corners of it ('corner-fired' or 'tangential'), and the characteristics of combustion will be very different in each case. The burners may be provided with individually controlled fuel and air supplies, or common control may be applied for all the burners, or they may be operated in groups, each group having dedicated and separately controlled supplies of fuel and air. Combustion of raw coal or other solid fuels such as municipal waste, clinical waste or refuse-derived fuels is often carried out in fluidised beds, or on stokers consisting of moving grates or platforms.

The methods of controlling these various arrangements are very different. With front-fired or opposed-fired boilers the temperature of the flue gases and the resulting heat transfer to the various banks of superheater tubes is adjusted by bringing burners into service or taking them out of service, and this may be done individually or in banks. This method provides a step-function type of control and fine adjustment of steam temperature is provided by spray-water attemporation.

Corner-fired (tangential) boilers are arranged in such a way that the burning fuel circulates around the furnace, forming a large swirling ball of burning fuel at the centre. With this type of boiler the manufacturers usually employ tilting mechanisms to direct the fireball to a higher or lower position within the furnace, and this has a significant effect on the temperature of the various banks of superheater tubes, and therefore on steam temperature. The maximum degree of tilt that is available within the basic design is typically $\pm 30°$, although the degree of movement employed in practice is usually restricted during commissioning.

The downside of tilting is that burners—with their fuel and air supplies, igniters, flame monitors etc.—are complex things, and tilting them requires very careful engineering if it is to be successful. Also, the tilting mechanisms must be rigorously maintained if they are to continue to operate effectively over any length of time.

The control systems that regulate burner tilting mechanisms must ensure that exactly the same degree of tilt is applied to the burners at all four corners of the furnace, since any misalignment will cause the fireball to circulate helically rather than as required.

3.2 The air and gas circuits

The combustion process requires the provision of fuel and air in the correct ratio to each other. This is known as the stoichiometic ratio, and under this condition enough air is provided to ensure complete combustion of all the fuel, with no surplus or deficit. However, this is a theoretical ideal, and practical considerations may necessitate operating at a fuel/air ratio that is different from the stoichiometric value. In addition, it must be understood that the efficiency of the combustion process will also be affected by the temperature of the air provided.

In the following sections we shall see how air is delivered to the furnace at the right conditions of flow and temperature, starting with the auxiliary plant that warms the air and moving on to the types of fan employed in the draught plant.

3.2.1 The air heater

In a simple-cycle plant, air is delivered to the boiler by one or more FD fans and the products of combustion are extracted from it by ID fans. Figure 3.1 shows this plant in a simplified form, and illustrates how the heat remaining in the exhaust gases leaving the furnace is used to warm the air being fed to the combustion chamber. This function is achieved in an air heater, which can be either regenerative, where an intermediate medium is used to transfer the heat from the exhaust gases to the incoming air, or recuperative, where a direct heat transfer is used across a dividing partition.

One variety of regenerative air heater is the Ljungström type, where metal plates mounted on a rotating frame are passed through the hot gases and then to the incoming air.

From a control engineer's point of view, an important consideration is the efficient combustion of the fuel, and here it is necessary to consider the

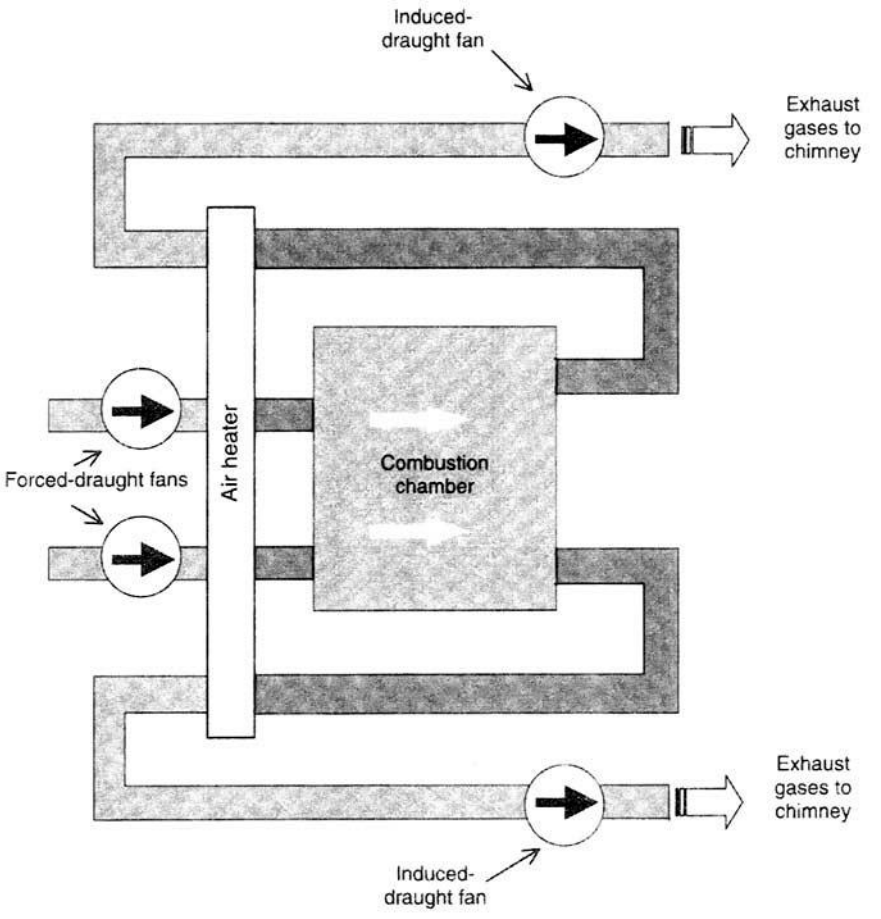


Figure 3.1 Draught-plant arrangement

losses and leakages that occur in an air heater. Figure 3.2 shows how various leakages occur in a typical air heater: across the circumferential, radial and axial seals, as well as at the hub. These leakages are minimised when the plant is first constructed, but become greater as wear occurs during prolonged usage. When the sheer physical size of the air heater is considered (Figure 3.3) it will be appreciated that these leakages can become significant.

3.2.2 Types of fan

In addition to the FD and ID fans mentioned above, another application for large fans in a power-station boiler is where it is necessary to overcome the resistance presented by plant in the path of the flue gases to the stack.

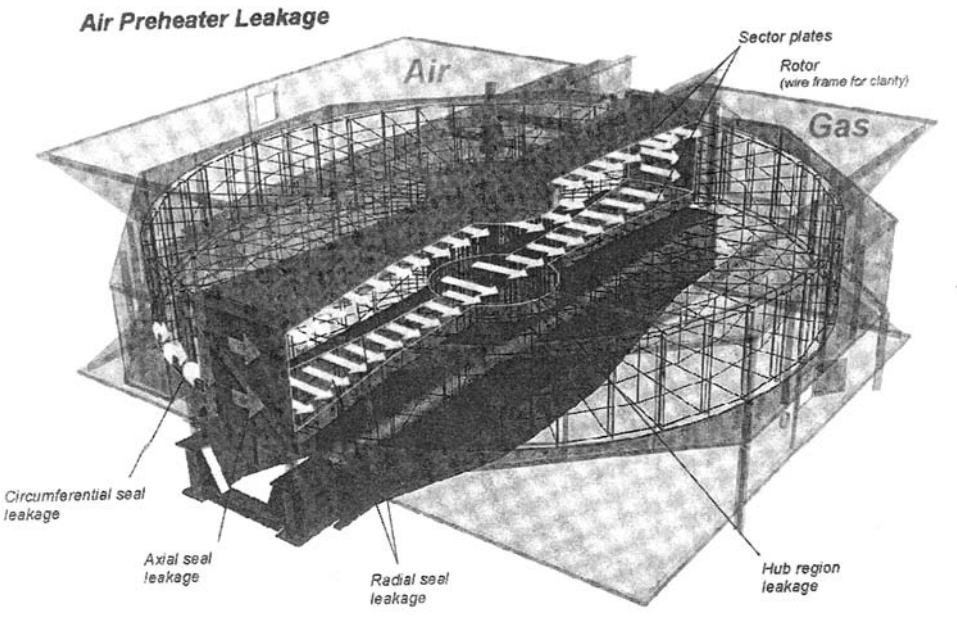


Figure 3.2 *Air heater leakage*
© Howden Sirocco Ltd. Reproduced with permission

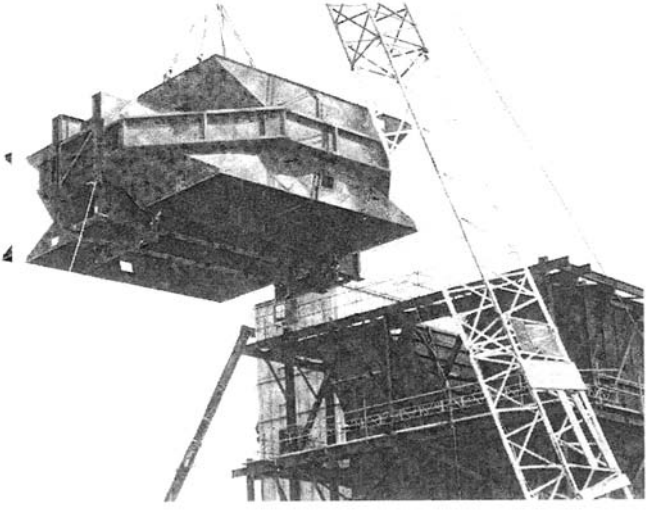


Figure 3.3 *An air heater being lifted into position*
© Howden Sirocco Ltd. Reproduced with permission

In some cases, environmental legislation has enforced the fitting of flue-gas desulphurisation equipment to an existing boiler. This involves the use of absorbers and/or bag filters, plus the attendant ducting, all of which present additional resistance to the flow of gases. In this case this resistance was not anticipated when the plant was originally designed, so it is necessary to fit additional fans to overcome the draught losses. These are called 'booster fans'.

Whatever their function, as far as the fans themselves are concerned, two types are found in power-station draught applications: centrifugal (Figure 3.4) and axial-flow (Figure 3.5). In the former, the blades are set radially on the drive shaft with the air or flue gas directed to the centre and driven outwards by centrifugal force. With axial-flow fans, the air or gas is drawn along the line of the shaft by the screw action of the blades. Whereas the blades of a centrifugal fan are fixed rigidly to the shaft, the pitch of axial-flow fan blades can be adjusted. This provides an efficient means of controlling the fan's throughput, but requires careful design of the associated control system because of a phenomenon known as 'stall', which will now be described.

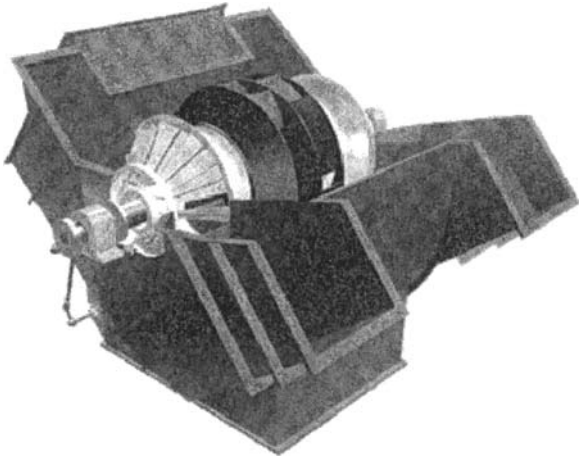


Figure 3.4 Centrifugal fan
© Howden Sirocco Ltd. Reproduced with permission

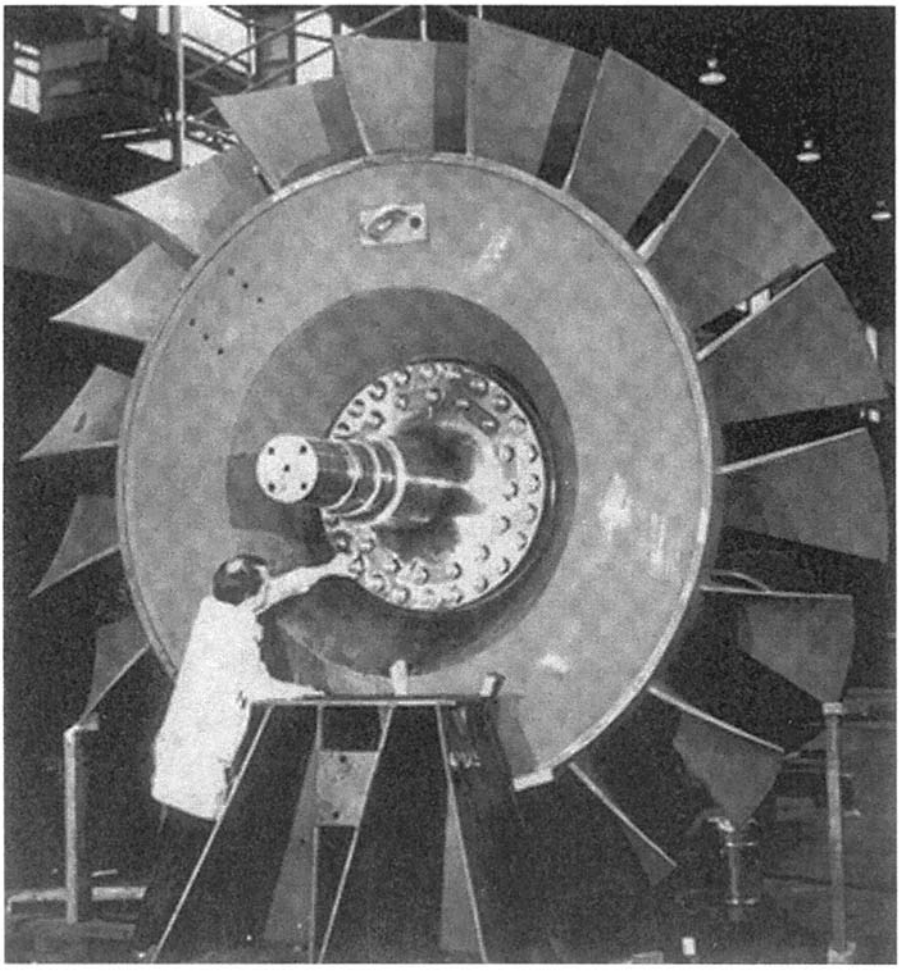


Figure 3.5 *Axial-flow fan*
© Howden Sirocco Ltd. Reproduced with permission

3.2.2.1 The stall condition

The angular relationship between the air flow impinging on the blade of a fan and the blade itself is known as the 'angle of attack'. In an axial-flow fan, when this angle exceeds a certain limit, the air flow over the blade separates from the surface and centrifugal force then throws the air outwards, towards the rim of the blades. This action causes a build-up of pressure at the blade tip, and this pressure increases until it can be relieved at the clearance between the tip and the casing. Under this condition the operation of the fan becomes unstable, vibration sets in and the flow starts to oscillate. The risk of stall increases if a fan is oversized or if the system resistance increases excessively.

For each setting of the blades there is a point on the fan characteristic beyond which stall will occur. If these points are linked, a 'stall line' is generated (Figure 3.6) and if this is built into the plant control system (DCS) it can be used to warn the operator that the condition is imminent and then to actively shift operation away from the danger region. The actual stall-line data for a given machine should be provided by the fan manufacturer.

3.2.2.2 Centrifugal-fan surge

The stall condition affects only axial-flow fans. However, centrifugal fans are subject to another form of instability. If they are operated near the peak of their pressure/flow curve a small movement either way can cause the pressure to increase or decrease unpredictably. The point at which this phenomenon occurs is known as the 'surge limit' and it is the minimum flow at which the fan operation is stable.

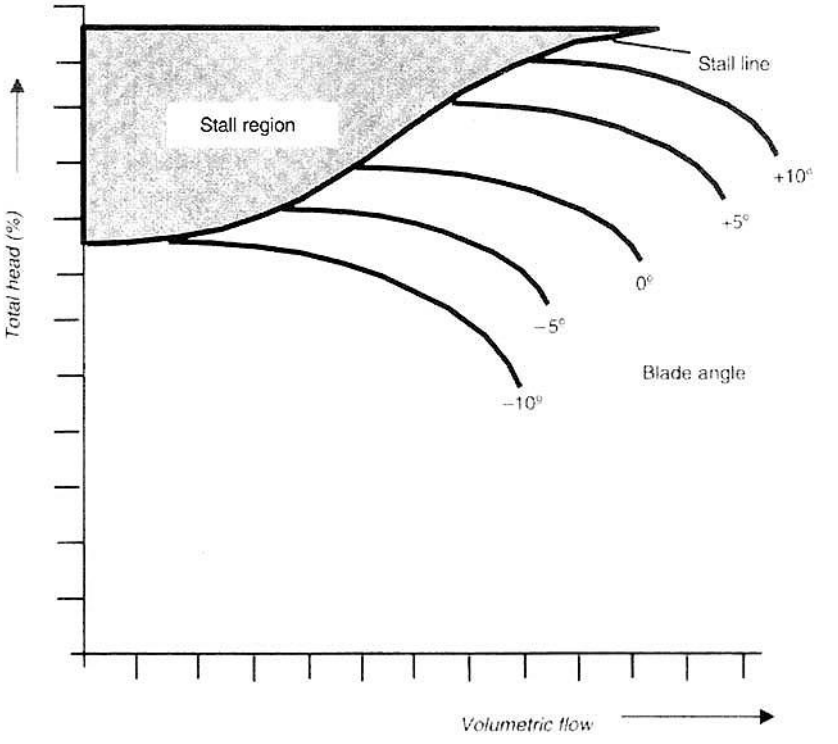


Figure 3.6 The stall line of an axial-flow fan

The system designer needs to be aware of the risk of surge occurring, since it may be necessary to adapt the control-system design. However, this is generally not a problem if the fan is properly designed in relation to the overall plant. During the initial design of the control system, dialogue with the process engineer or boiler designer will show whether or not surge protection will be required.

3.2.3 Final elements for draught control

Reference has already been made to the use of pitch-control in axial-flow fans to regulate the throughput of the machine. Other means of controlling flow are dampers, vanes or speed adjustment. Each of these devices has its own characteristics, advantages and disadvantages, and the selection of the controlling device which is to be used in a given application will be a trade-off between the technical features and the cost.

3.2.3.1 Types of damper

The simplest form of damper consists of a hinged plate that is pivoted at the centre so that it can be opened or closed across the duct. This provides a form of draught control but it is not very linear and it is most effective only near the closed position. Once such a damper is more than about 40–60% open it can provide very little additional control. Another form of damper comprises a set of linked blades across the duct (like a Venetian blind). Such multibladed dampers are naturally more expensive and more complex to maintain than single-bladed versions, but they offer better linearity of control over a wider range of operation.

The task of designing a control system for optimum performance over the widest dynamic range will be simplified if the relationship between the controller output signal and the resultant flow is linear. Although it is possible to provide the required characterisation within the control system, this will usually only be effective under automatic control. Under manual control a severely nonlinear characteristic can make it difficult for the operator to achieve precise adjustment.

It is possible to linearise the command–flow relationship under both manual and automatic control by the design of the mechanical linkage between the actuator and the damper. However, this requires careful design of the mechanical assemblies and these days it is generally considered simpler to build the required characterisation into the DCS. This approach provides a partial answer, but it should not be forgotten that such a solution is only effective under automatic control.

3.2.3.2 Vane control

The second form of control is by the adjustment of vanes at the fan inlet. These vanes are clearly visible near the centre of Figure 3.7 (which shows a centrifugal fan during manufacture). Such vanes are operated via a complex linkage which rotates all the vanes through the same angle in response to the command signal from the DCS.

3.2.3.3 Variable-speed drives

Finally, control of fan throughput can be achieved by the use of variable-speed motors (or drives). These may involve the use of electronic controllers which alter the speed of the driving motor in response to demand signals from the DCS or they can be hydraulic couplings or variable-speed gearboxes, either of which allows a fixed-speed motor to drive the fan at the desired speed. Variable speed drives offer significant advantages in that they allow the fan to operate at the optimum speed for the required throughput of air or gas, whereas dampers or vanes control the flow by restricting it, which means that the fan is attempting to deliver more flow than is required.

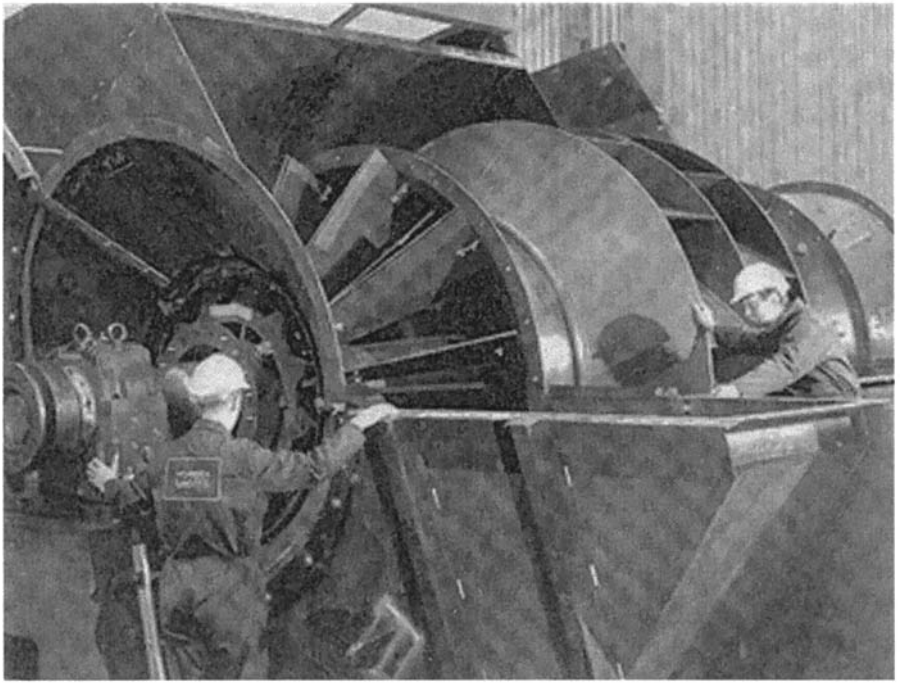


Figure 3.7 A centrifugal fan during manufacture
© Howden Sirocco Ltd. Reproduced with permission

3.3 Fuel systems

Fossil fuels that are burned in boilers can be used in solid, liquid or gaseous form, or a mixture of these. Naturally, the handling systems for these types of fuels differ widely. Moreover, the variety of fuels being burned is enormous. Solid fuels encompass a wide spectrum of coals as well as wood, the waste products of industrial processes, municipal and clinical waste and refuse-derived fuels. (The last are produced by shredding or grinding domestic, commercial and industrial waste material.) Liquid fuels can be heavy or light oil, or the products of industrial plant. Gas can be natural or manufactured, or the by-product of refineries.

Each of these fuels requires specialised handling and treatment, and the control and instrumentation has to be appropriate to the fuel and the plant that processes it.

3.3.1 Coal firing (pulverised fuel)

Although coal can be burned in solid form on grates, it is more usual to break it up before feeding it to the combustion chamber. The treatment depends on the nature of the coal. Some coals lend themselves to being ground down to a very fine powder (called pulverised fuel (PF)) which is then carried to the burners by a stream of air. Other coals are fed to impact mills which use flails or hammers to break up the material before it is propelled to the burners by an air stream. The type of mill to be used on a particular plant will be determined by the process engineers and it is the task of the control engineer to provide a system which is appropriate. To do this it is necessary to have some understanding of how the relevant type of mill operates.

Various types of pulverised-fuel mill will be encountered, but two are most commonly used: the pressurised vertical-spindle ball mill and the horizontal-tube mill.

3.3.1.1 Vertical-spindle ball mills

Figure 3.8 shows the operating principle of a typical ball mill, such as the Babcock 'E' mill. In this device, the coal that is discharged from the storage hoppers is fed down a central chute onto a table where it is crushed by rotating steel balls. Air is blown into the crushed coal and carries it, via adjustable classifier blades, to the PF pipes that transport it to the burners.

The air that carries the fine particles of coal to the burners is supplied from a fan called a 'primary-air fan'. This delivers air to the mill, which therefore operates under a pressure which is slightly positive with respect

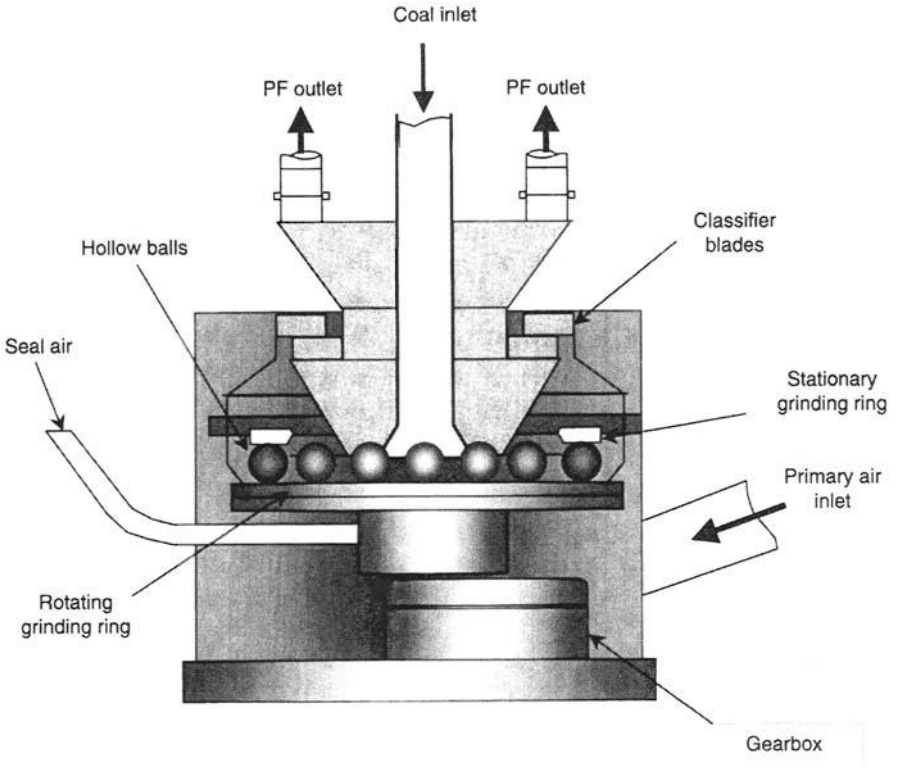


Figure 3.8 Pressurised ball mill

to the atmosphere outside. Because of this and because of its other constructional features, this type of mill is properly called a 'vertical-spindle, pressurised ball mill'. The air-supply system for this type of mill is discussed in more detail in Section 3.3.1.3.

3.3.1.2 Horizontal tube mills

In a tube mill (Figure 3.9) the coal is fed into a cage that rotates about a horizontal axis, at a speed of 18 to 35 rpm. This cage contains a charge of forged-steel or cast alloy balls (each of which is between 25 mm and 100 mm in diameter) which are carried up the sides of the cage by the rotation, until they eventually cascade down to the bottom, only to be carried up again. The coal is pulverised by a combination of the impact with these balls, attrition of adjacent particles and crushing between the balls and the cage and between one ball and another.

In this type of mill the crushed mixture is drawn out of the cage by a fan, which is called an exhauster. Because of this configuration, the tube

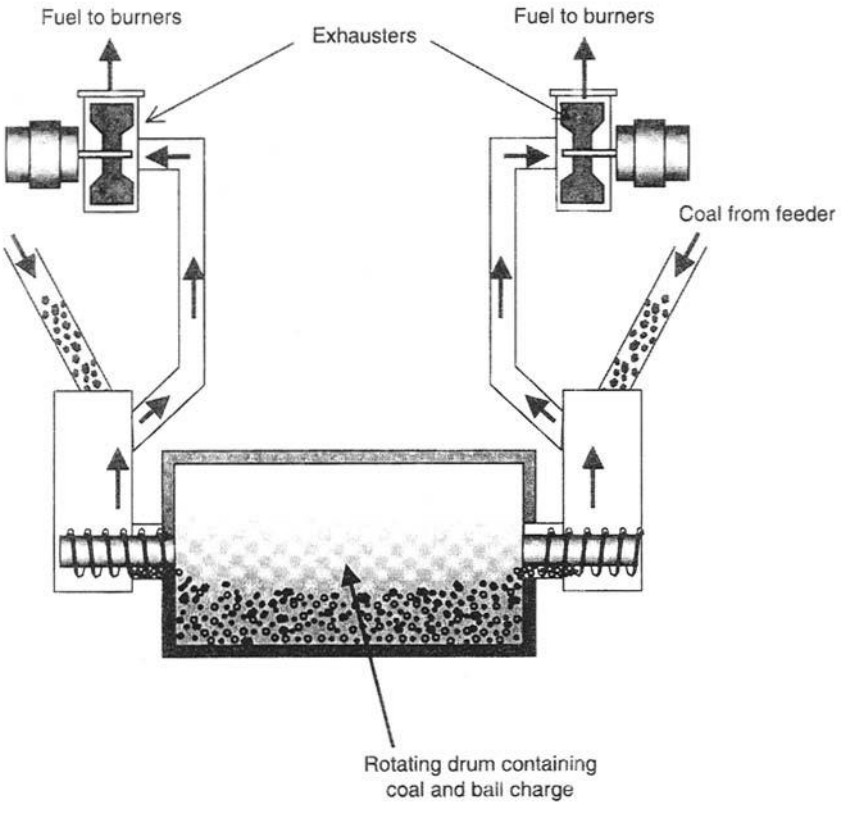


Figure 3.9 Horizontal tube mill

mill runs under a negative pressure, which prevents the fine coal dust from escaping (as it tends to do with a pressurised mill). However, the exhausters have to handle the dirty and abrasive mixture of coal and air that comes through the mill and they therefore require more frequent maintenance than the fans of a pressurised ball mill, whose function is merely to transport air from the atmosphere to the mill.

3.3.1.3 Air supply systems for mills

As stated above, the crushed coal in a pressurised ball mill is propelled to the burners by a stream of warm air. Figure 3.10 shows the arrangement for doing this: cool air and heated air are mixed to achieve the desired temperature. This temperature has to be high enough to partially dry the coal, but it must not be so high that the coal could overheat (with the risk of the coal/air mixture igniting inside the mill or even exploding while it is being crushed). The warm air is then fed to the mill (or a group of mills) by

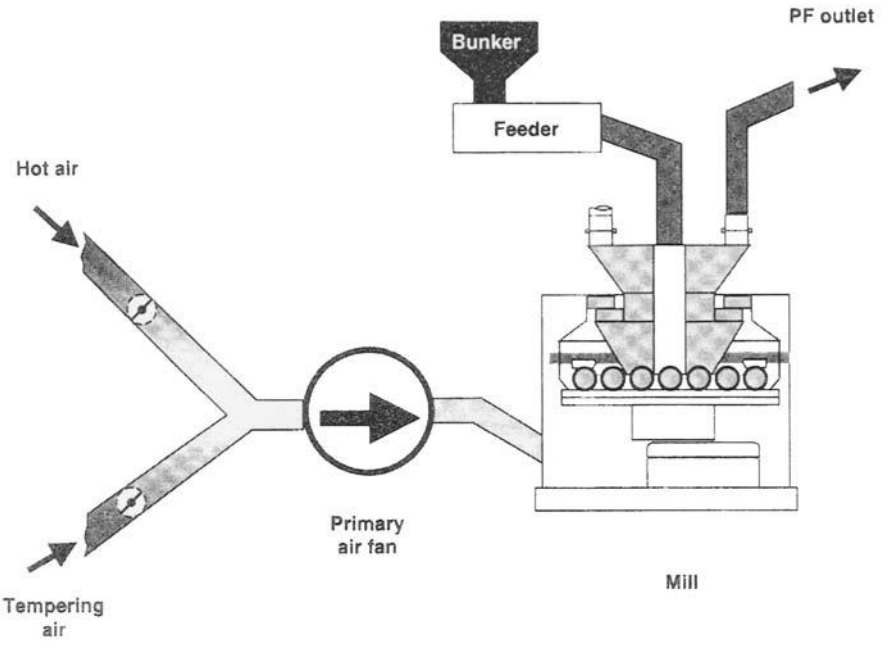


Figure 3.10 Primary air fan system for a ball mill

means of yet another fan, called a 'primary air fan'. It should be noted that the cooler of the two air streams is commonly referred to as 'tempering air' since, because it is obtained from the FD fan exhaust it may already be slightly warm, and its function is to temper the mixture.

Figure 3.11 shows the system that is used with a tube mill. Here, hot air and cold air are again mixed to obtain the correct temperature for the air stream but, because the mill in this case operates under suction conditions a primary air fan is not needed, and the cold air is obtained directly from the atmosphere. The warmed air mixture is again fed to the mill as 'primary air' but in addition a stream of hot air is fed to the feeder for transportation and drying purposes.

3.3.2 Oil-firing systems

In comparison with coal, oil involves the use of much less capital plant. On the face of it, it would appear that all that is required is to extract the oil from its storage tank and pump it to the burners. But in reality life is more complicated than that!

Proper ignition of oil depends on the fuel being broken into small droplets (atomised) and mixed with air. The atomisation may be achieved

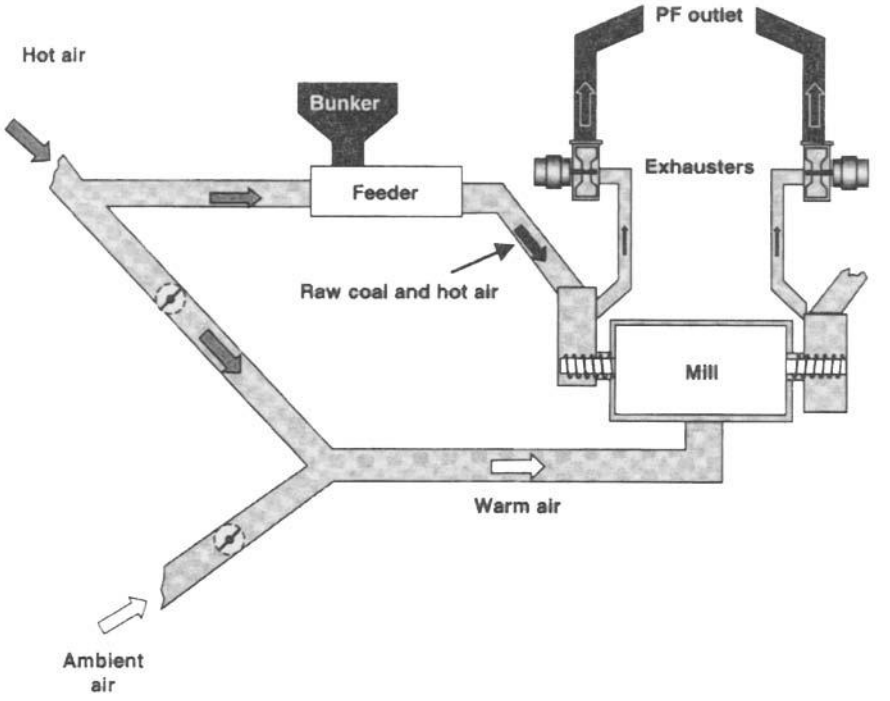


Figure 3.11 Suction mill air supply system

by expelling the oil through a small nozzle (a 'pressure jet'), or it may be achieved by the use of compressed air or steam.

The fuel oil itself may be light (such as diesel oil or gas oil), or it may be extremely viscous and tar-like (heavy fuel oil, commonly 'Bunker C'). The handling system must therefore be designed to be appropriate to the nature of the liquid. With the heavier grades of oil, prewarming is necessary, and to prevent it cooling and thickening the fuel is continually circulated to the burners via a recirculation system (shown schematically in Figure 3.12). The latter process is sometimes referred to as 'spill-back'. When a burner is not firing, the oil circulates through the pipework right up to the shut-off valve, which is mounted as close to the oil gun as possible.

From the point of the C&I engineer, the control systems involved with oil firing may include any or all of the following: controlling the temperature of the fuel, the pressure of the atomising medium, and the equalisation of the fuel pressures at various levels on the burner front.

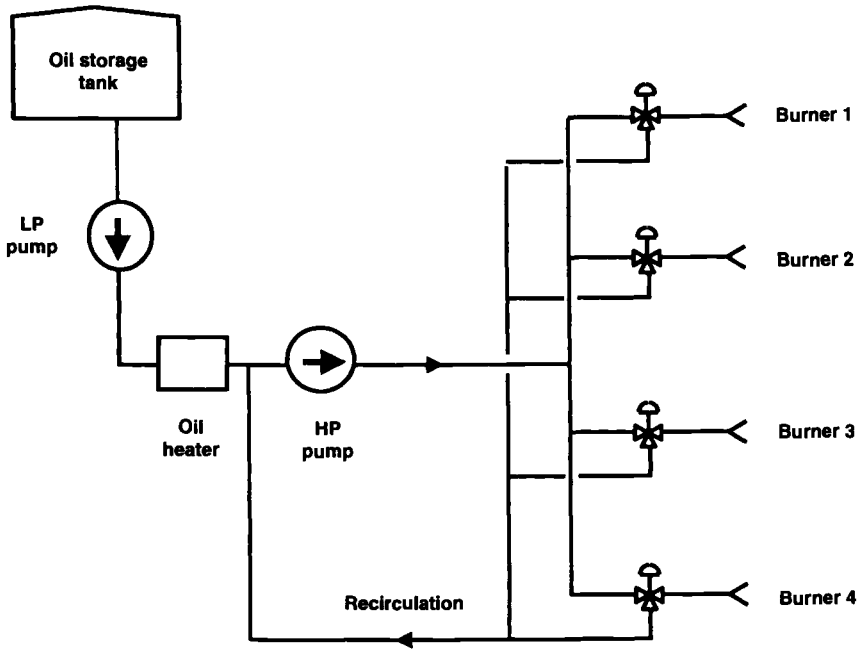


Figure 3.12 Simplified oil pumping, heating and recirculation system

3.3.3 Gas-firing systems

Although inherently simpler than either oil or coal-fired systems, gas-fired boilers have their own complexities. Any escape of gas, particularly into confined areas, presents considerable hazards, and great care must therefore be taken to guard against leakage, for example, from flanges and through valves. But natural gas is colourless, and any escape will therefore be invisible. Also, it is not safe to rely on odour to detect leakages. By the time an odour has been detected sufficient gas may have already escaped to present a hazard. It is therefore necessary for gas-leak detectors to be fitted along the inlet pipework wherever leakage could occur, and to connect these to a comprehensive, central alarm system.

It is also necessary to prevent gas from seeping into the combustion chamber through leaking valves. If gas does enter undetected into the furnace during a shut-down period, it could collect in sufficient quantities to be ignited either by an accidental spark or when a burner is ignited. The resulting explosion would almost certainly cause major damage and could endanger lives. (It should be noted that this risk is present with propane igniters such as those used with fuels other than oil.)

Protection against leakage into the furnace through the fuel-supply valves is achieved by the use of 'double-block-and-bleed' valve assemblies which provide a secure seal between the gas inlet and the furnace. The operation of this system (see Figure 3.13) is that before a burner is ignited both block valves are closed and the vent is open. In this condition any gas which may occupy the volume between the two block valves is vented to a safe place and it can therefore never develop enough pressure to leak past the second block valve. When start-up of the burner is required, a sequence of operations opens the block valves in such a way that gas is admitted to the burner and ignited safely.

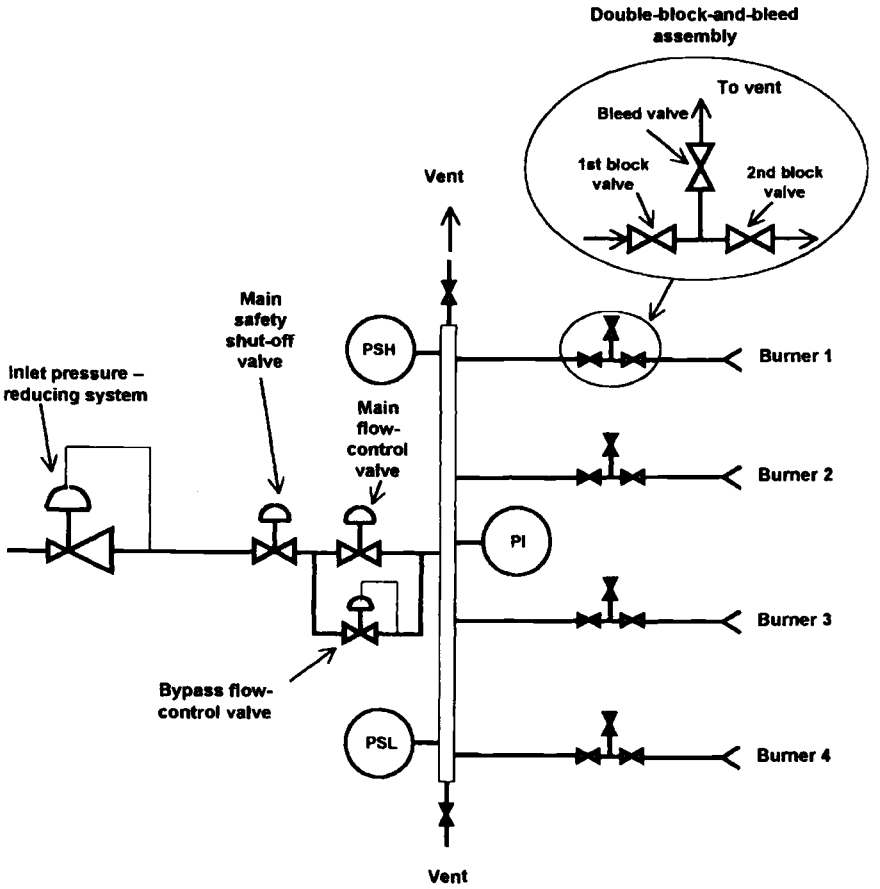


Figure 3.13 Simplified schematic of gas-firing system

3.3.4 Waste-to-energy plants

There has been a steady development of plants that incinerate waste material of various types and use the heat thus produced to generate electricity. Early units suffered from the unpredictable nature of the waste material and the severe corrosion resulting from the release of acidic compounds during the combustion process. But the problems have been largely overcome through the application of improved combustion systems and by better knowledge of the materials used in the construction of the plant.

Waste material may be obtained from any of several sources, including the following:

- municipal;
- industrial;
- clinical;
- agricultural.

The material may be burned after very basic treatment (shredding etc.) or it may be processed in some way, in which case the end result is termed refuse-derived fuel (RDF).

Several types of waste-to-energy plant are in existence, and we shall look at one of them, so that its nature and characteristics can be appreciated. Other plants will differ in their construction or technology, but from an operational point of view their fundamental characteristics will probably be quite similar to those described below.

3.3.4.1 The bubbling fluidised-bed boiler

Figure 3.14 shows the principles of a waste-to-energy plant based on the use of a bubbling fluidised-bed boiler. First, the incoming waste is sorted to remove oversized, bulky or dangerous material. The remainder is then carried by a system of conveyors to a hammer mill where it is broken down until only manageable fragments remain. After a separator has removed incombustible magnetic items, the waste is held in a storage building, from where it is removed as required by a screw conveyor and transferred via another conveyor to the boiler. Immediately before entering the boiler, nonferrous metals are removed by a separator.

The boiler itself comprises a volume of sand which is kept in a fluidised state by jets of air. A portion of dolomite is added to the sand to assist in the reduction of corrosion and to reduce any tendency of the sand and fuel to coalesce (a process known as 'slagging'). After the sand/dolomite mixture has been heated by a system of start-up burners, combustible

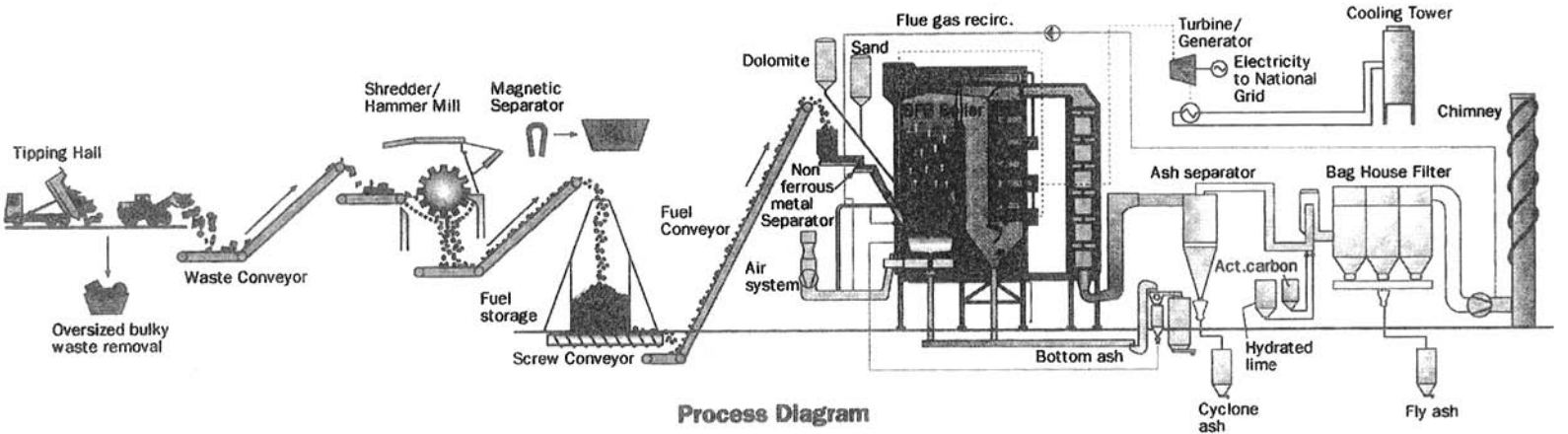


Figure 3.14 Waste-to-energy plant based on bubbling fluidised-bed boiler

© Kvaerner 1999 Reproduced by permission

waste material added to it ignites. The heat released is used to generate steam in a way that is similar to conventional boilers such as those described in Chapter 2.

3.4 Igniter systems

Whatever the main fuel of the boiler may be, it is necessary to provide some means of igniting it. A variety of igniters are used, but most modern systems comprise a means of generating a high-energy electric spark which lights a gas or light-oil supply which in turn lights the main fuel.

In addition to igniting the fuel, the igniter may sometimes be used to ensure that the fuel remains alight under conditions where it may otherwise be extinguished. This is referred to as providing 'support' for the main burner.

Like many aspects of power-station burner operations, the requirements for igniters are defined in standards such as those developed by the National Fire Protection Association (e.g. NFPA 8502:95 [1]). In these standards, igniters are divided into three categories each of which is defined in detail. In essence, the three classes have the following characteristics.

Class 1: An igniter providing sufficient energy to raise the temperature of the fuel and air mixture above the minimum ignition temperature, and to support combustion, under any burner light-off or operating conditions. Such igniters generally have a capacity of more than 10% of the full-load capacity of the main burner that it is igniting. This class is also referred to as a 'Continuous Igniter'.

Class 2: (also referred to as an 'intermittent igniter'): Capable of lighting the fuel *only under a defined range of light-off conditions.* Such igniters have a capacity generally between 4% and 10% of the full-load burner input and may also be used to support combustion of the fuel at low loads or under a defined range of adverse operating conditions.

Class 3: Small igniters, generally applied to gas or oil burners. These igniters are capable of lighting the fuel *only under a defined range of conditions and may not be used for support purposes.* Two types of Class 3 igniter are defined: interrupted igniters (not usually exceeding 4% of the main burner fuel input energy), whose operation is automatically stopped when a set time has expired after the first ignition; and direct electric igniters which have enough energy to ignite the main fuel.

The type of igniter in use will define the methods of operation of the burner and the sequences that are to be employed in the associated burner-management system.

3.5 Burner-management systems

Safe operation of the burner and its associated igniter must be ensured and in most cases this requires the use of a sophisticated burner-management system (BMS). In outline, these systems include a means of monitoring the presence of the flame and a reliable method and procedure for operating the associated fuel valves in a sequence that provides safe ignition at start-up and safe shut-down, either in the event of a fault or in response to an operator command.

The procedure for lighting a burner depends, first, on checking that it is safe to light it at all. This means that, if no other burner is firing, confirmation has been received that any flammable mixtures have been exhausted from the furnace by means of a purge. Such a purge involves the operation of FD and ID fans for a defined time, so that a certain volume of air has passed through the furnace. (In a coal-fired boiler the flow rate through the furnace must be at least 40% of the full-load volumetric air flow.)

Once confirmation has been received that the furnace purge is complete (or if other burners are already firing), ignition of the burner will depend on the successful operation of some form of igniter or pilot and, once the main burner has been successfully lit, its operation must be continuously monitored, because an extinguished flame may mean that unburned fuel is being injected into the combustion chamber. If such fuel is subsequently ignited it may explode.

Once a burner has ignited, the BMS must ensure that safe operation continues, and if any hazard arises the system must shut off the burner, and if necessary, trip the entire boiler.

On shut-down of a burner, steps must be taken to ensure that any unburned fuel is cleared from the pipework. This procedure is known as scavenging, and in an oil burner it may involve blowing compressed air or steam through the pipework and burner passages. Such procedures are defined in codes such as NFPA 8502-95.

Each component of the BMS is vital to the safety of the plant and to the reliability of its operation, but the most onerous responsibility rests with the flame detector: an electronic device which is required to operate in close proximity to high-energy spark ignition systems, and in conditions of extreme heat and dirt. Moreover, it must provide a reliable indication of

the presence or absence of a particular flame in the presence of many others and it must discriminate between the energy of the flame and high levels of radiant energy from hot refractory materials and pipes. The sighting of the flame may also be affected by changes in flame pattern over a wide range of operating conditions, and it may also be obscured by swirling smoke, steam or dust.

Safe operation of the boiler depends on proper design of the BMS, including the flame scanner, and on careful siting of the scanner so that it provides reliable and unambiguous detection of the relevant flame under all operational conditions. After installation, the system can be expected to perform safely and reliably *only* if constant and meticulous attention is paid to maintenance. This important matter is all too often ignored, and the inevitable result is that the system malfunctions, leading to failure to ignite the fuel, which may in turn delay start-up of the boiler. In the extreme, malfunctions could even endanger the safety of the plant if they result in fuel being admitted to the combustion chamber without being properly ignited. A properly designed BMS will not allow this to happen, but if repeated malfunctions occur it is not unknown for operators to ignore the warning signs and even to override safety systems. In such cases it is usual to blame the BMS and/or the flame monitors, which could be fully functional if they were not misused or badly maintained. This important subject is discussed in greater depth in Chapter 5.

3.6 Gas turbines in combined-cycle applications

In the combined-cycle plant, the heat used for boiling the water and superheating the steam is obtained from the exhaust of a gas turbine, as described in Chapter 2. In such plant, unless supplementary firing is used, the combustion process occurs entirely in the gas turbine. Where supplementary firing is used the relevant control systems take on many of the characteristics of the oil- or gas-firing systems discussed earlier in the present chapter.

3.7 Summary

So far, we have looked at the operation of the boiler and studied in outline the boiler's steam, water and gas circuits, and all the major items of plant required for their operation. With this understanding we can now look at the control and instrumentation systems associated with the plant. This survey will be structured in much the same way as the preceding chapters, starting with an overview of an important fundamental: the

method by which the demand for steam, heat or electrical power is obtained. Afterwards, we shall see how this demand is transmitted to all the relevant sections of the plant so that the requirements are properly and safely addressed.

3.8 References

1 NFPA 8502-95: Standard for the prevention of furnace explosions/ implosions in multiple burner boilers. National Fire Protection Association, Batterymarch Park, Quincy, MA, USA, 1995